

Exercise Sheet 1

Discrete Mathematics, 2023.09.14

1. Check whether the following sentences are propositions. For every sentence which is a proposition, judge whether they are true or false. Please give reasons for your answers.

- $5 + 7 \leq 10$
- "Is Beijing the capital of China?"
- $x + 3 = 7$
- $x + 1 > x$ for any real number x
- For all real numbers x , there is a real number y such that $xy = 1$
- If $5 + 7 > 10$, then $5 > 10$

2. Let p and q be the propositions:

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- It is below freezing and snowing.
- It is below freezing but not snowing.
- It is not below freezing and it is not snowing.
- It is either snowing or below freezing (or both).
- If it is below freezing, it is also snowing.
- That it is below freezing is necessary and sufficient for it to be snowing.

3. Construct a truth table for each of these compound propositions (If the truth value of a propositional variable does not matter, you can use "-" to simplify the case for the propositional variable.):

a) $p \rightarrow \neg p$

b) $p \leftrightarrow \neg p$

c) $p \oplus (p \vee q)$

d) $(p \wedge q) \rightarrow (p \vee q)$

e) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

f) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

4. Construct a truth table for $((p \rightarrow q) \rightarrow r) \rightarrow s$ (If the truth value of a propositional variable does not matter, you can use "-" to simplify the case for the propositional variable.)

Answers

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1.
 - yes,false, $5 + 7 = 12 > 0$.
 - no,it is not a statement.
 - no,the value of x is unknown.
 - yes,true.
 - yes,false,because when $x=0$,there is not a real number y such that $xy=1$.
 - yes,false,because " $5 + 7 > 10$ " is true but " $5 > 10$ " is false.

2. $p \wedge q$.
 $p \wedge (\neg q)$.
 $\neg p \wedge \neg q$.
 $p \vee q$.
 $p \rightarrow q$.
 $p \leftrightarrow q$.

3. a)

p	$\neg p$	$p \rightarrow \neg p$
T	F	F
F	T	T

- b)

p	$\neg p$	$p \leftrightarrow \neg p$
T	F	F
F	T	F

- c)

p	q	$p \vee q$	$p \oplus (p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	F

- d)

p	q	$p \wedge q$	$p \vee q$	$((p \wedge q) \rightarrow (p \vee q))$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

- e)

p	q	$\neg p$	$q \rightarrow \neg p$	$p \leftrightarrow q$	$(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$
T	T	F	F	T	F
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T

f)

p	q	$\neg q$	$p \leftrightarrow q$	$p \leftrightarrow \neg q$	$(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	F	T	T
F	F	T	T	F	T

4.

p	q	r	s	$p \rightarrow q$	$(p \rightarrow q) \rightarrow r$	$((p \rightarrow q) \rightarrow r) \rightarrow s$
T	T	T	T	T	T	T
T	T	T	F	T	T	F
T	T	F	T	T	F	T
T	T	F	F	T	F	T
T	F	T	T	F	T	T
T	F	T	F	F	T	F
T	F	F	T	F	T	T
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	T	F	T	T	F
F	T	F	T	T	F	T
F	T	F	F	T	F	T
F	F	T	T	T	T	T
F	F	T	F	T	T	F
F	F	F	T	T	F	T
F	F	F	F	T	F	T